FARM MANAGEMENT SYSTEM

User Story:

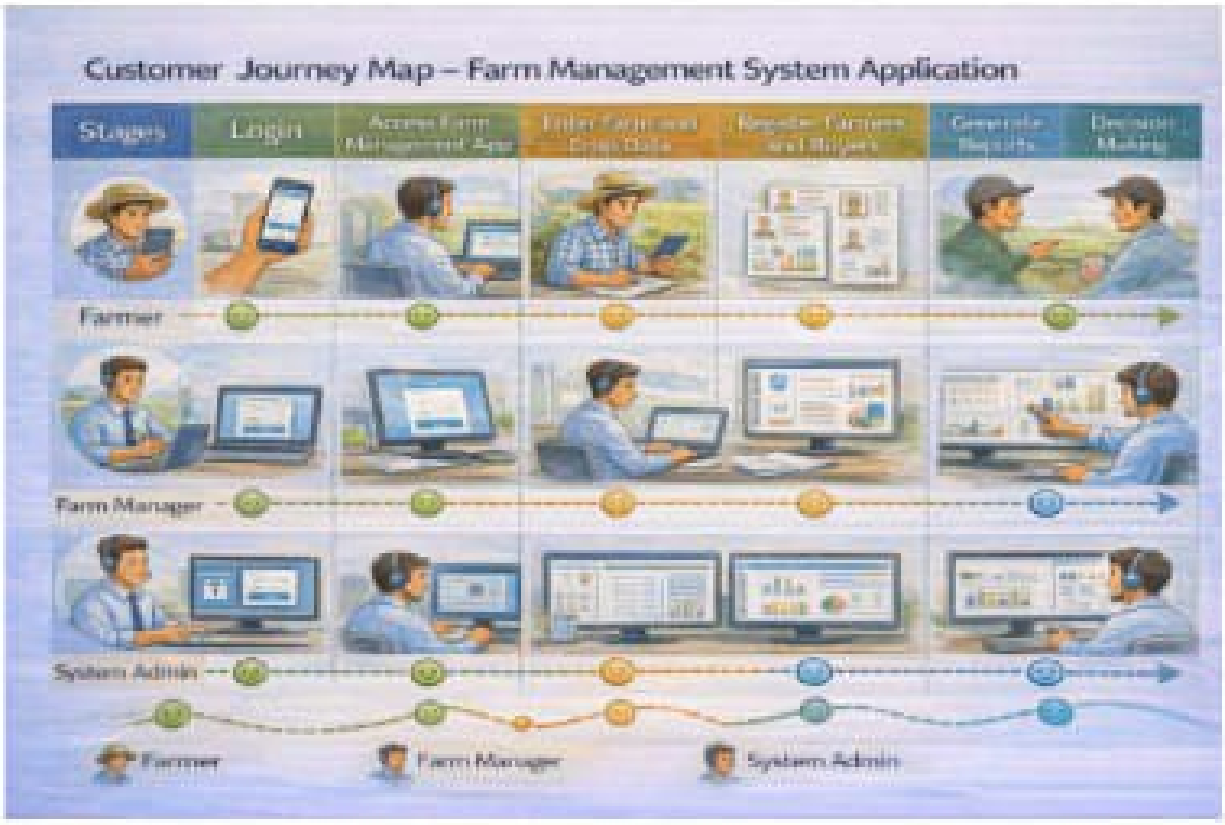
In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation

Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.es
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards

7. Click Close to close the preview window, Click Save.

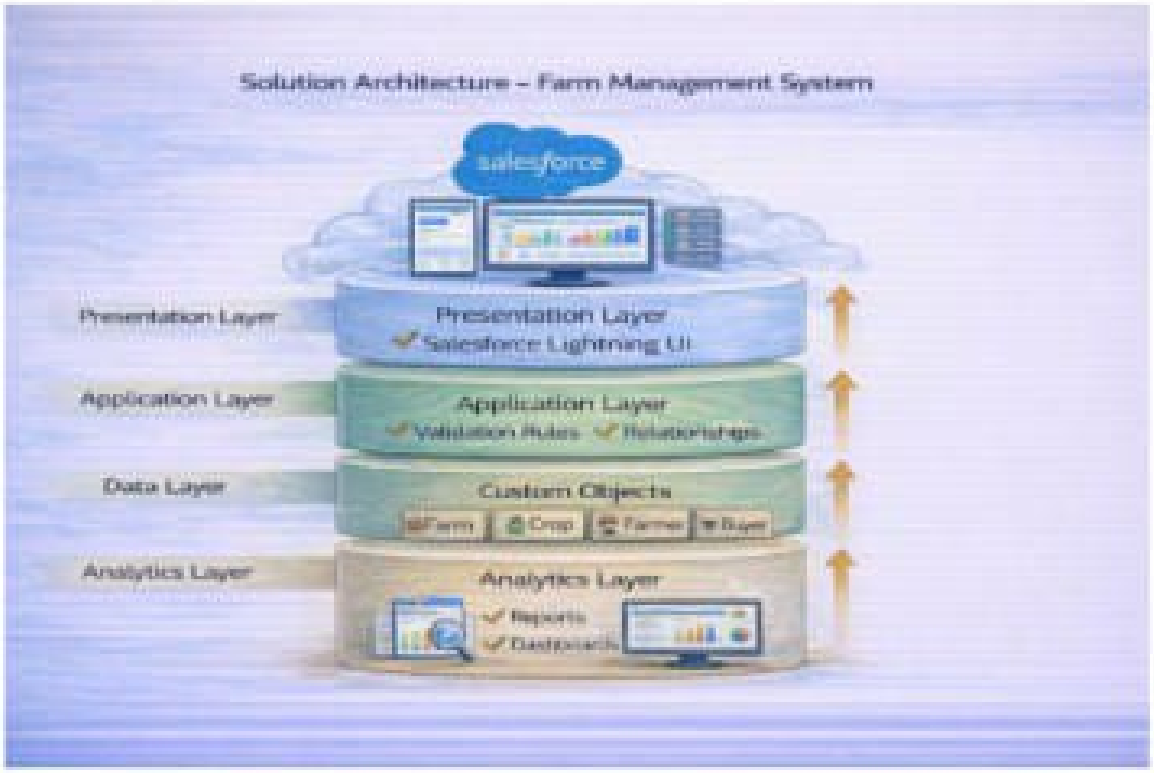
Activity 4:

Creation of fields for the Buyer objects:

1.Select Data type as a Phone,Enter a Field Label(Contact),Save.
2.Select Data type as an Email,Enter Field Label(Email), Next save.
3.Select Data type Text area,Enter Field Label(Address),Save it.
4.Select Data type Picklist, Field Label(Buyer Type),Enter values (Wholesale,Retail),Click Save.

Milestone 6- Profile:

What is a profile?
A profile is a group/collection of settings and permissions that define what a user can do in



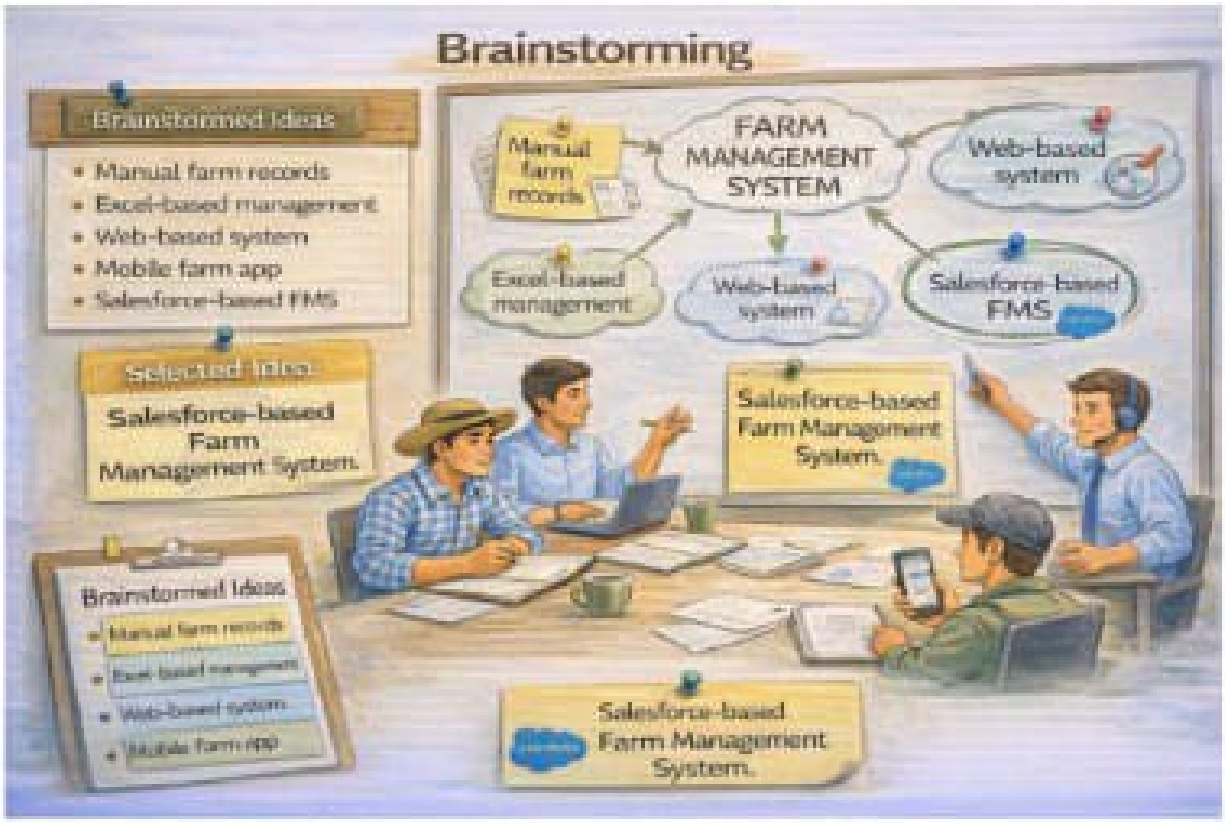
5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making

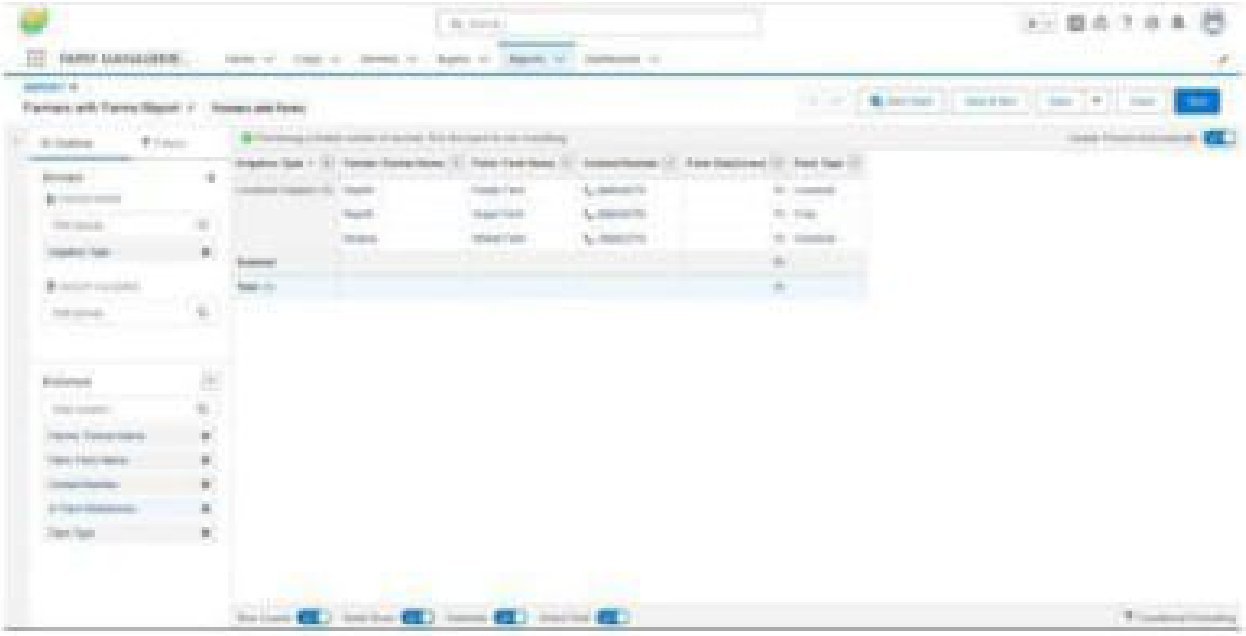
Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)
2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)
3.Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter field label (size)
5.Create a Master Detail relationShip with Farmer Object.

Activity 2:
Creation of fields for the Crop objects:

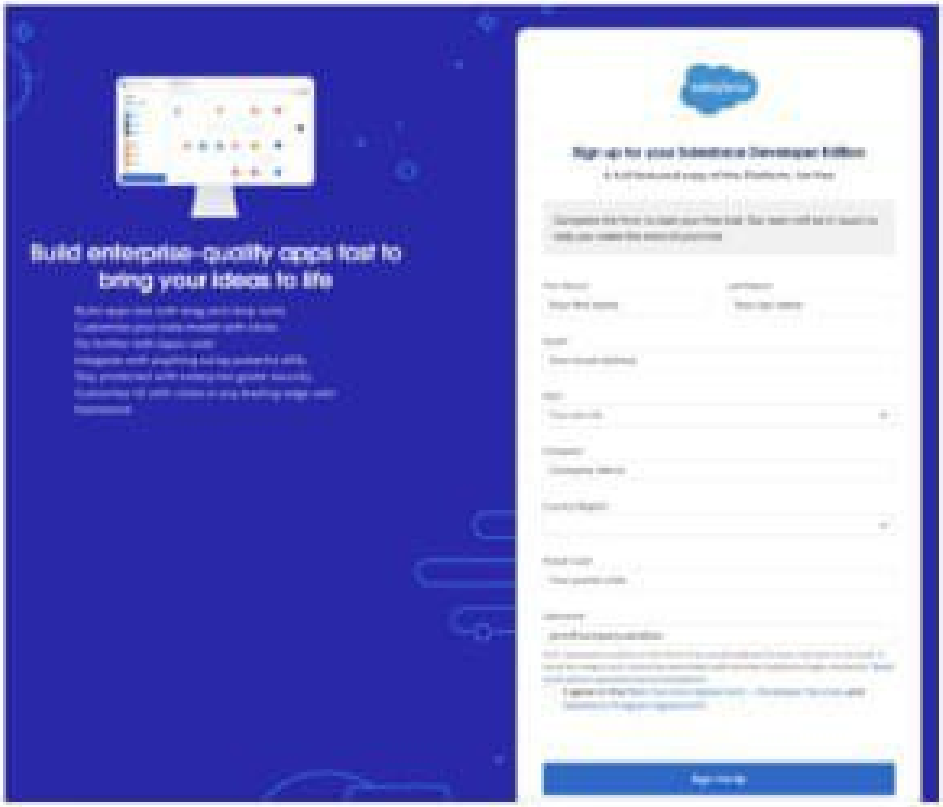
1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save. 2.Select Data type as date ,Enter Field label (Planting Date),then save .
3.Again we select data type as a date,Enter field labe (Harvest date),save
4.Next We Create Lookup relationship on crop object realted to Farm Object.
5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.
6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.
7.Now We create Formula Field For That Select data Type As a Formula,Click Next.

3) In the details section select the option start report.
4) In Outline Click on Column Select Which Field you want to display in the Report.
5) Click Group,Select Which Fields You Want to Group.
6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



Activity1:
Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.
1. Search Developer.salesforce.com/signup
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce area group of tabs that help the application function by working together asa unit. It has a name,a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app

3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:
Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?
Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.
Types of sharing rules.
•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

Create a sharing rule
1) Goto Sharing Settings, which can be found under the Quick Find section.
2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
3) Add the label of the sharing rule you want to make.
4) Select your rule type based on the criteria.
5) Select the field can join immediately check field from the Crop object.
6) Select the Crop as equal and value is True.
7) And in selecting the users to share with the section select roles and in that select Farm Manager.
8) And in the section of select the level of access for the users give the access Read/Write.
9) And save the rule.

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Problem Statements – Farm Management System

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1		Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3		Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

c. Plural Label: Farmers
d. Record Name: Farmer Name e. Select the data type as "Text".
f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:
1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
a. On the Custom Object Definition page, create the object as follows:
b. Label: Buyer
c. Plural Label: Buyers
d. Record Name: Buyer Name
e. Select the data type as "Text".
f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
i. Click Save.

Milestone 3 -
Tab: Tab

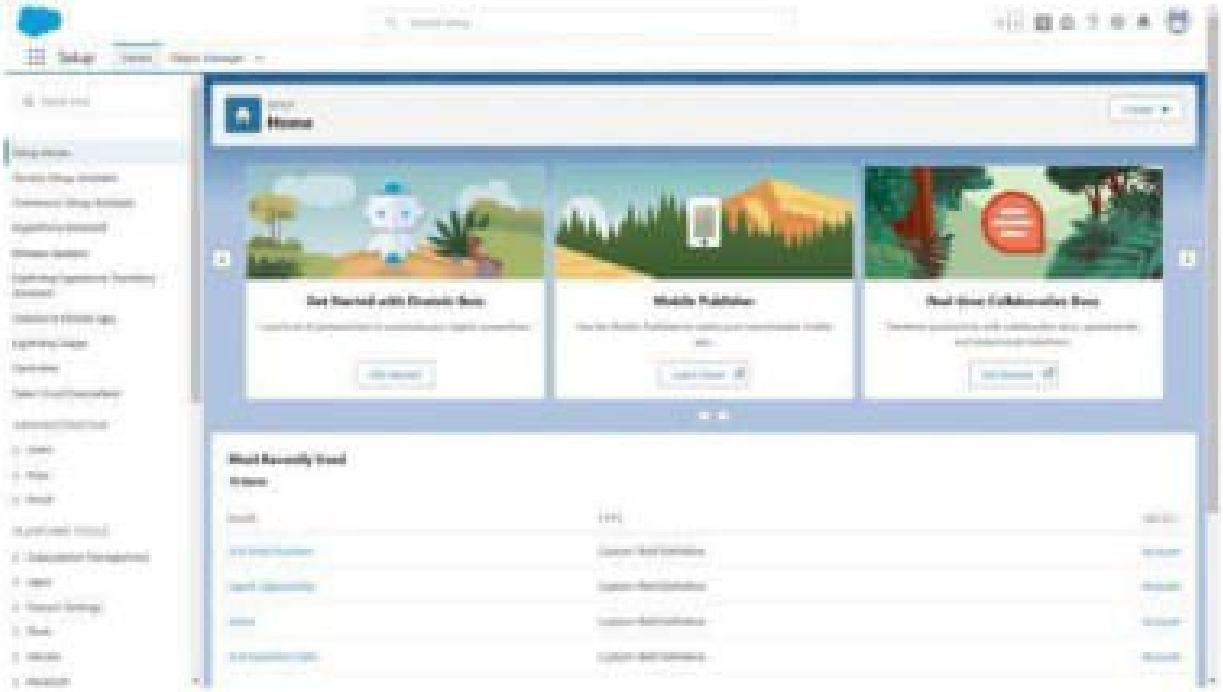
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

Now create a custom tab.

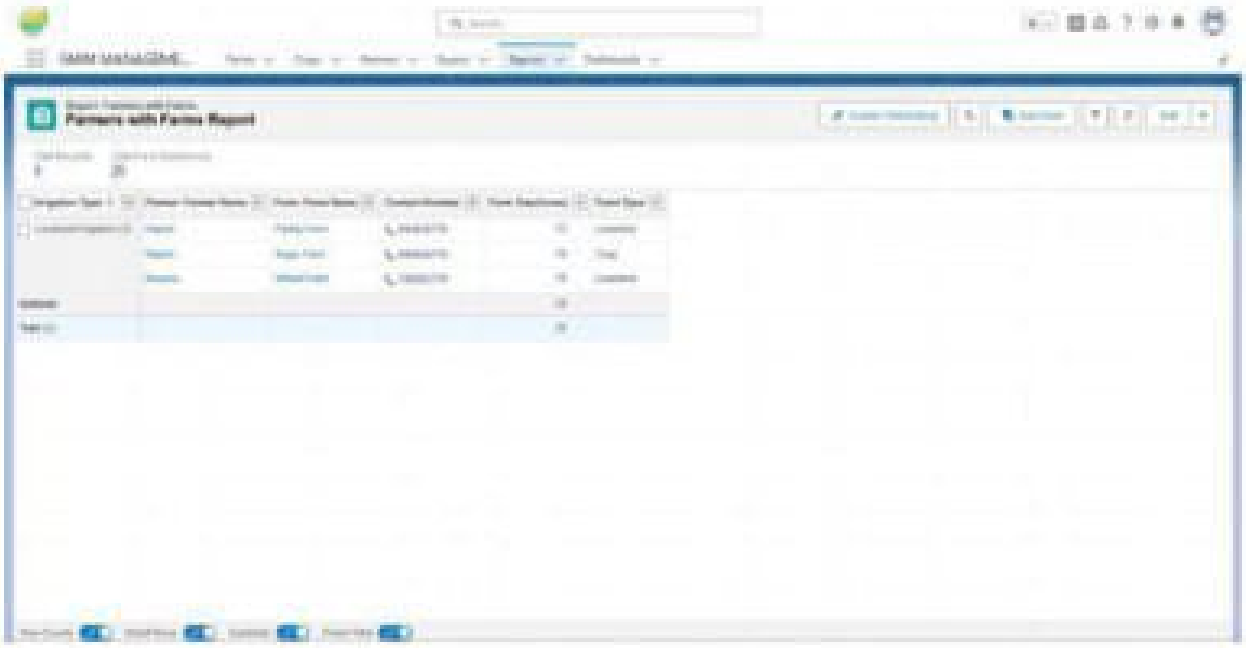
6. By using username and password you can into the Salesforce org. The homepage will appear below



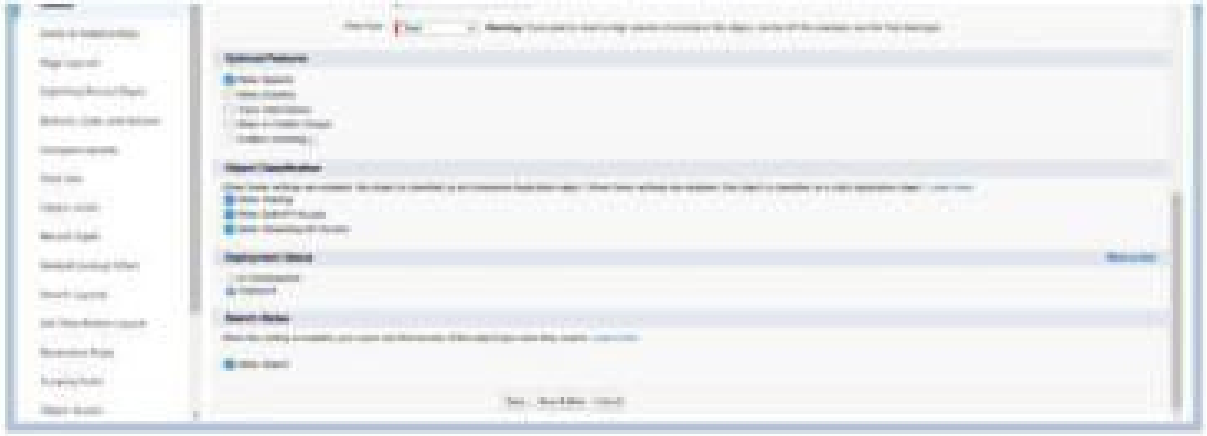
Milestone-13: View Report and Dashboard:

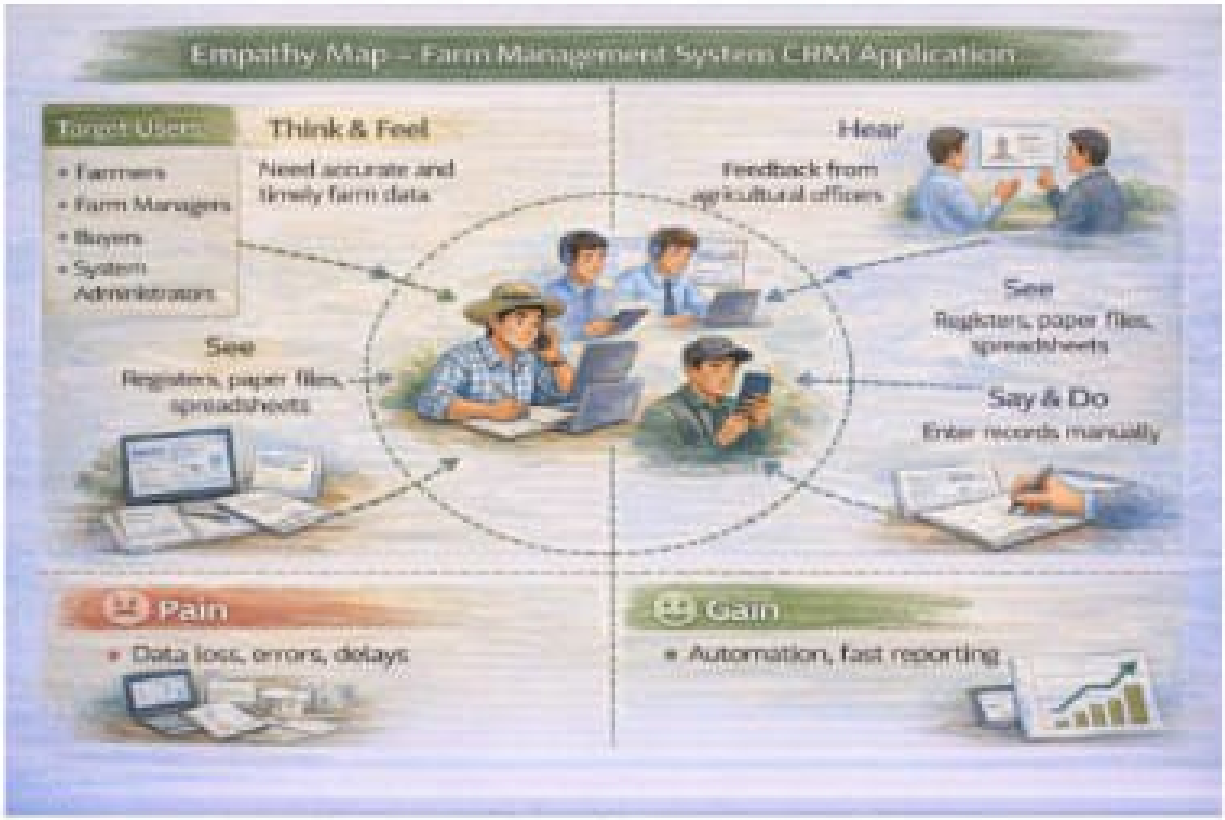
Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".





2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

salesforce. A profile controls "Object permissions, Field permissions, User permissions, Tab
settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.
A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.

Activity 3:
Creation of fields for the Farmer objects:

1.Select Data type Phone,Enter field label(Contact Number),next Save.
2.Select Data type Picklist,Enter Field Label(State),Enter values (Telangana,Andhra Pradesh,Tamilnadu),save.
3.Choose Data type Picklist,Enter Field Label (District),Enter Values of district,next save.
4.Again select Data type as Picklist,Enter Field Label (Village),Enter Values of villages,next save.

Creation of field dependency :

1.Now create Field Dependency For District field,Open Object Manager,Choose Farmer object ,select Fields&Relationships,On Right side We have Field Dependency option,select that option.
2.Click new, Select Controlling Field as State,Dependent Field As District,Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5.Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.

6.Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

Search Setup

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

Crop

- Details
- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types

Fields & Relationships

11 Items, Sorted by Field Label

Quick Find | New | Deleted Fields | Field Dependencies | Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Actual Yield	Actual_Yield__c	Currency(18, 0)		
Created By	CreatedById	Lookup(User)		
Crop Name	Crop_Name__c	Text(100)		
Crop Name	Name	Text(80)		✓
Expected Yield per Acre	Expected_Yield_per_Acre__c	Formula (Currency)		

Search...

Farm Management System

Farms | Crops | Farmers | Buyers | Reports | Dashboards

Report: Crops with Farm

Crops with Farm Report

Enable Field Editing | Add Chart | Edit

Total Records	Total Actual Yield	Total Total Acres Harvested	Total Expected Yield per Acre
3	$600,000	30	$60,000.00

Farm: Farm Name	Sum of Actual Yield	Sum of Total Acres Harvested	Sum of Expected Yield per Acre	Record Count
Gayathri Farm	$500,000	25	$40,000.00	2
Shivashankar Farm	$100,000	5	$20,000.00	1
Total	$600,000	30	$60,000.00	3

Row Counts | Detail Rows | Subtotals | Grand Total

31°C Mostly cloudy

Search

ENG IN

18:20 10-03-2026

Setup | Home | Object Manager

Sharing

Security

Guest User Sharing Rule Access Report

Sharing Settings

Didn't find what you're looking for? Try using Global Search.

SETUP
Sharing Settings

Sharing Settings

Help for this Page

This page displays your organization's sharing settings. These settings specify the level of access your users have to each others' data. Go to Background Jobs to monitor the progress of a change to an organization-wide default or a parallel sharing recalculation.

One or more sharing operations has been initiated. See below for additional details. Certain operations may not be available.

Manage sharing settings for: All Objects

Disable External Sharing Model

Default Sharing Settings

Organization-Wide Defaults

Edit

Organization-Wide Defaults Help

Object	Default Internal Access	Default External Access	Grant Access Using Hierarchies
Lead	Public Read/Write/Transfer	Private	✓
Account and Contract	Public Read/Write	Private	✓
Contact	Controlled by Parent	Controlled by Parent	✓
Order	Controlled by Parent	Controlled by Parent	✓
Asset	Controlled by Parent	Controlled by Parent	✓

Developer Edition

Search...

Farm Management System

Farms | Crops | Farmers | Buyers | Reports | Dashboards

Report: Crops with Farm

Crops with Farm Report

Enable Field Editing | Add Chart | Edit

Total Records	Total Actual Yield	Total Total Acres Harvested	Total Expected Yield per Acre
3	$600,000	30	$60,000.00

Farm: Farm Name	Crop: Crop Name	Actual Yield	Total Acres Harvested	Expected Yield per Acre	Farm: Farmer
Gayathri Farm (2)	Rice	$300,000	15	$20,000.00	Gayathri
	Cotton	$200,000	10	$20,000.00	Gayathri
Subtotal		$500,000	25	$40,000.00	
Shivashankar Farm (1)	Coconut	$100,000	5	$20,000.00	Shivashankar
Subtotal		$100,000	5	$20,000.00	
Total (3)		$600,000	30	$60,000.00	

Row Counts | Detail Rows | Subtotals | Grand Total

31°C Search ENG 15:16

Search Setup

Setup | Home | Object Manager

pro

Hyperforce Assistant

- Users
  - Profiles
- Data
  - Mass Transfer Approval Requests
- Feature Settings
  - Approval Settings
  - Data.com
    - Prospector Preferences
    - Prospector Users
  - Functions
  - Marketing
    - Lead Processes
  - Sales
    - Products

SETUP

Profiles

Profile

Sales Department

Help for this Page

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

Login IP Ranges [0] | Enabled Apex Class Access [0] | Enabled External Data Source Access [0] | Enabled Named Credential Access [0] | Enabled External Credential Principal Access [0] | Enabled Custom Metadata Type Access [0] | Enabled Custom Setting Definitions Access [0] | Enabled Flow Access [0] | Enabled Service Presence Status Access [0] | Enabled Custom Permissions [0]

Profile Detail

Edit | Clone | Delete | View Users

Name	Sales Department		
User License	Analytics Cloud Integration User	Custom Profile	✓
Description			
Created By	M.Venishree,S.Anusiya P.Lathika,A.Swetha,S.Archana, 3/10/2026, 1:03 AM	Modified By	M.Venishree,S.Anusiya P.Lathika,A.Swetha,S.Archana, 3/10/2026, 1:03 AM

Page Layouts

Standard Object Layouts

Global	Global Layout [View Assignment]	Location Group Assignment	Location Group Assignment Layout [View Assignment]
Email Application	Not Assigned [View Assignment]	Macro	Macro Layout [View Assignment]
Home Page Layout	Home Page Default [View Assignment]	Object Milestone	Object Milestone Layout [View Assignment]

30°C Sunny | Search | ENG IN | 13:34 10-03-2026

Developer Edition

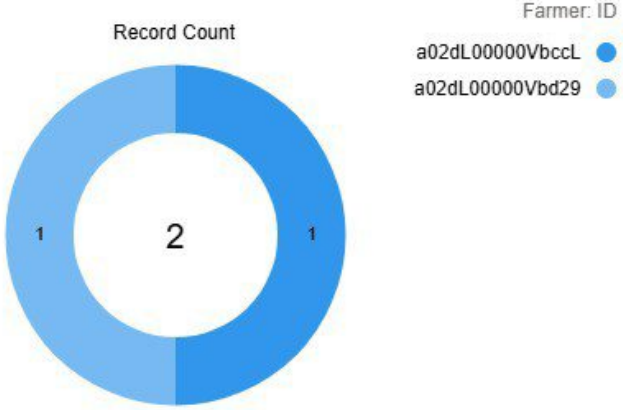
Search...

Farm Management System

Farms | Crops | Farmers | Buyers | Reports | Dashboards

View Report (Crops with Farm Repo... As of Mar 10, 2026, 2:52 AM

Farmers with Farms Report



View Report (Farmers with Farms R... As of Mar 10, 2026, 2:52 AM

31°C Mostly cloudy

Search

ENG IN

18:06 10-03-2026

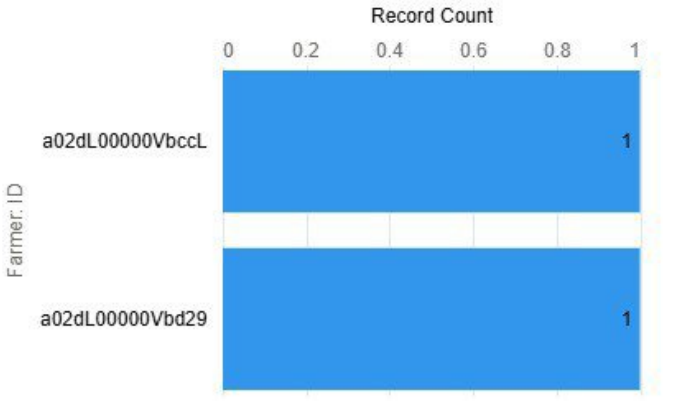
Dashboard

Farm Management Dashboard

As of Mar 10, 2026, 2:52 AM · Viewing as M.Venishree,S.Anusiya P.Lathika,A.Swetha,S.Archana

Refresh | Edit | Subscribe

Farmers with Farms Report

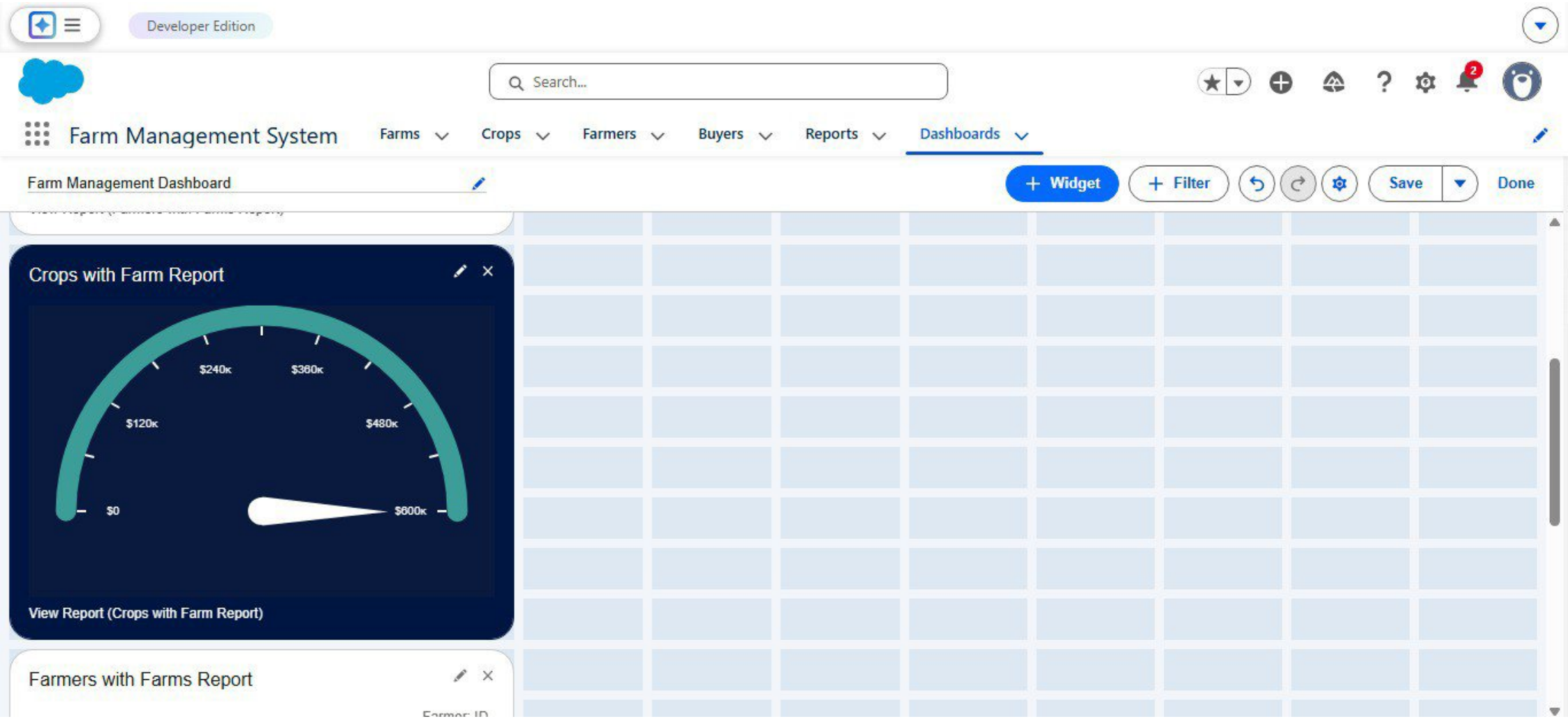


View Report (Farmers with Farms R...

As of Mar 10, 2026, 2:52 AM

Crops with Farm Report

31°C
Mostly cloudy
Search
ENG
IN
18:22
10-03-2026


Developer Edition
Search...
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
Farm Management Dashboard
+ Widget
+ Filter
Save
Done
Crops with Farm Report
$0
$120K
$240K
$360K
$480K
$600K
View Report (Crops with Farm Report)
Farmers with Farms Report

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thank You